

3.5 Overview

An application for managing and optimising the distribution and management of water resources is known as a "water supply application." It makes use of technology and data to track, examine, and enhance several elements of water supply systems. Here is a description of an application for water supply. Data collection is a crucial part of a water supply application. It collects information from a variety of devices, including sensors, metres, weather stations, and user inputs. Information on water use, water quality, infrastructure condition, environmental variables, and other pertinent elements are all included in this data. Preprocessing is done to the data after it has been collected. The data must be cleaned, missing values must be handled, outliers must be eliminated, and data consistency must be ensured. In order to extract meaningful information from the data, feature engineering is another step in the process of preparing data. To extract meaning from the gathered data, the application makes use of a variety of analytical methodologies. In order to do this, it may be necessary to examine past consumption trends, spot anomalies or leaks in the water distribution system, forecast water demand, and optimise water allocation depending on supply and demand circumstances. A water supply application must provide visualisation. It use graphs, charts, maps, and other visual representations to clearly and simply display the analysed data. Operators, managers, and decision-makers in the water supply industry can use visualisations to make wise choices and carry out the necessary activities. Automation and control systems are frequently used in water supply applications to improve how well water distribution networks function. This can entail putting algorithms into practise for real-time monitoring, managing pressure levels, controlling water flow, and synchronising pumps and valves to ensure effective water supply.